What is Java?

Java is a **high-level, object-oriented programming language** used to build applications like websites, mobile apps, desktop software, and enterprise systems.

Simple Explanation

Java is like a **universal language for computers**.

- You write code **once**
- It can run on **any device** (Windows, Mac, Linux)

This concept is called:
"Write Once, Run Anywhere" (WORA)

Key Components

1. JVM (Java Virtual Machine)

- Acts like a **translator**
- Converts Java code into machine-understandable format
- Makes Java platform independent

Example:
Your .class file runs inside JVM

2. JDK (Java Development Kit)

- Full package for **developers**
- Includes:
    - JVM
    - Compiler (javac)
    - Development Tools

Used to **write + compile + run Java programs**

Java Development Tools

In **Java**, **development tools** are the software that helps you **write, compile, run, debug, and manage Java applications**.

Summary

- JDK = Core development kit
- IDE = Coding environment (IntelliJ IDEA, Eclipse IDE, NetBeans)
- Build Tools = Used to **automate build process** (compile, test, package), (Apache Maven, Gradle)
- Git = Version control
- Server = Run apps(Apache Tomcat, JBoss)

3. JRE (Java Runtime Environment)

- Used to **run Java programs**
- Includes:
    - JVM
    - Libraries

End users only need JRE

Standard Libraries

In Java, Standard Libraries are pre-written classes and packages provided by Java to help you perform common tasks easily.

Simple Explanation

Standard Libraries = **Ready-made tools**

Why Use Standard Libraries?

- Saves time
- Reduces code length
- Reliable & tested
- Improves productivity

Common Java Standard Library Packages

java.lang (Default package) – String, Math, System

java.util – ArrayList, HashMap, Date

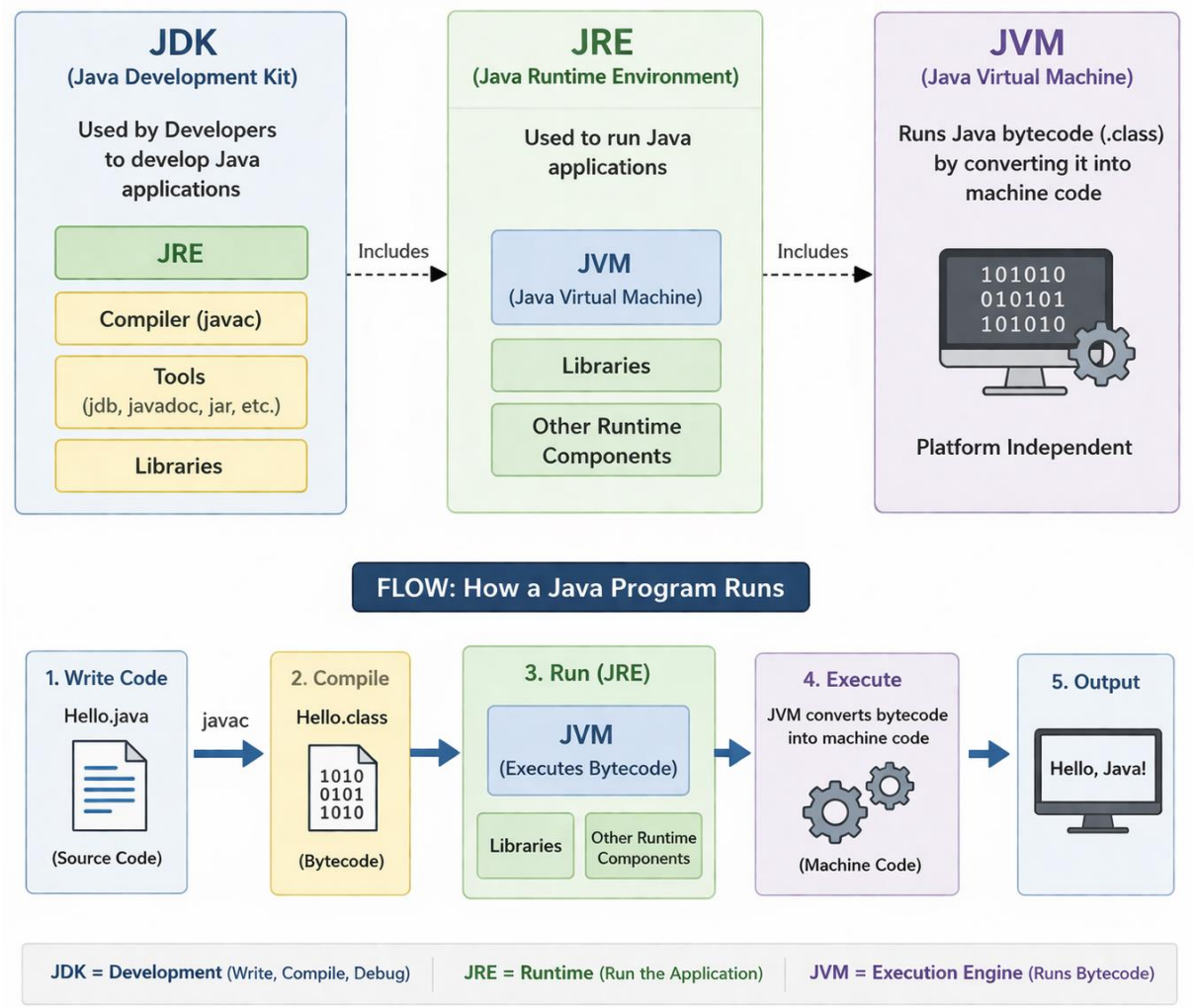
java.io – File, BufferedReader

java.sql - Database connectivity

java.time - Date & Time API

How Java Works (Flow)

Java Code (.java)
↓
Compiler (javac)
↓
Bytecode (.class)
↓
JVM converts
↓
Machine Code
↓
Output

Key Differences

Feature	Bytecode	Machine Code
Generated By	Java Compiler (javac)	JVM / Compiler
File Type	.class	Binary (no extension usually)
Readable By	JVM	CPU (Processor)
Platform	Platform Independent	Platform Dependent
Execution	Needs JVM	Directly executed
Format	Intermediate Instructions	Binary (0s & 1s)